Solutions — 5 Mock Exams · Data Science Introduction

This file contains the answer keys and short rationales for all five practice exams. For single-best-answer exams, one option is correct (1 point). For multiple-response exams, all listed options must be ticked (2 points; one mistake = 1 point; two or more mistakes = 0 points). Try each exam closed-book first, then check here.

Comprehensive Mock Exam — Solutions & Answer Key (Single best answer (tick one) · 1 pt/question · 90 points)

Q	Topic	Answer	Rationale
1	1. VSM vs. System Dynamics	**B**	VSM (Beer) = recursive structure for viability; SD (Forrester) = simulation of behaviour over time. Both rely on feedback.
2	1. VSM vs. System Dynamics	**B**	Both are feedback-based systems-thinking approaches; B is the incorrect statement.
3	1. VSM vs. System Dynamics	**B**	VSM = recursive control for viability (Beer); System Dynamics = simulation of dynamic behaviour over time (Forrester).
4	2. Ashby Adaptation	**B**	Adaptation = aligning internal variety with environmental variety, via amplification or attenuation.
5	2. Ashby Adaptation	**D**	Reducing one's own variety is not the only (or correct) response; D is incorrect.
6	2. Ashby Adaptation	**B**	Adaptation = match variety via amplification (own variety up) or attenuation (environment variety down).
7	3. Data Science Distinction	**D**	Big Data = data traits/infrastructure (4 Vs); Data Science = methods + theory + application across disciplines.
8	3. Data Science Distinction	**A**	Big Data and Data Science are not the same; A is incorrect.
9	3. Data Science Distinction	**C**	Big Data (data/infrastructure) and Data Science (broad discipline) are not interchangeable; C is incorrect.
10	4. Real-Time Adaptive Models	**C**	Real-time adaptive models update continuously from streaming feedback rather than being fixed once.
11	4. Real-Time Adaptive Models	**C**	Adaptive models must keep updating; C is incorrect.
12	4. Real-Time Adaptive Models	**B**	Continuous updating from a data stream is online/real-time adaptive modeling.
13	5. Overfitting	**A**	A large train-test gap with very high training accuracy indicates overfitting (high variance).
14	5. Overfitting	**C**	Increasing complexity worsens overfitting; C is incorrect.
15	5. Overfitting	**D**	Overfitting is reduced by simplification, more data, regularization and proper validation - not more complexity.
16	6. Visualization	**B**	A truncated axis is technically true but not fair - it can mislead the reader.
17	6. Visualization	**B**	Interpreting visualizations requires visual literacy; B is incorrect.
18	6. Visualization	**A**	A polished look does not make a chart objective; design choices can bias it. A is incorrect.
19	7. Complexity Examples	**C**	Complex CPS data is large, high-frequency, heterogeneous and non-IID.

20	7. Complexity Examples	**D**	Complexity is not the same as mere complicatedness; D is incorrect.
21	7. Complexity Examples	**A**	Non-IID = dependencies across time/context; sensor streams are the classic example.
22	8. Class Weighting	**D**	Class weighting adjusts the loss to value the minority class more, leaving the data unchanged.
23	8. Class Weighting	**B**	Class weighting does not add/remove rows; B is incorrect.
24	8. Class Weighting	**D**	Class weighting changes the loss (data unchanged); oversampling adds minority rows.
25	9. Target-Oriented Action	**D**	Target-oriented action begins with the goal and designs everything around it.
26	9. Target-Oriented Action	**D**	Starting data-first and ignoring the goal is the failure mode; D is incorrect.
27	9. Target-Oriented Action	**B**	Starting data-first and hoping for a goal is the anti-pattern target-orientation avoids; B is the answer.
28	10. Explainability of Models	**B**	XAI explains the 'why' behind decisions, which builds trust and supports responsible use.
29	10. Explainability of Models	**C**	Opaque models are not ideal for high-stakes decisions; C is incorrect.
30	10. Explainability of Models	**B**	XAI exposes the reasons behind decisions, supporting trust and accountability where stakes are high.
31	11. Ashby Control	**C**	To control a system, the controller's variety must match (>=) the system's variety.
32	11. Ashby Control	**D**	A low-variety controller cannot manage a high-variety system; D is incorrect.
33	11. Ashby Control	**C**	Requisite variety: the controller's variety must be >= the system's variety.
34	12. Viable System Model	**D**	The VSM is a recursive management structure with anti-oscillatory feedback ensuring viability.
35	12. Viable System Model	**C**	The VSM is not a static, feedback-free chart; C is incorrect.
36	12. Viable System Model	**A**	The VSM is recursive and feedback-based, not a linear feedback-free plan; A is incorrect.
37	13. Class Imbalance	**C**	Under imbalance, a trivial majority-class predictor looks accurate yet misses the minority class.
38	13. Class Imbalance	**B**	High accuracy can hide poor minority-class performance; B is incorrect.
39	13. Class Imbalance	**B**	With heavy imbalance, accuracy is misleading; the model ignores the minority class (recall ~ 0).
40	14. Uncertainty Scenarios	**B**	Uncertainty = unknown outcomes despite data; D=complexity, A=ambiguity, C=volatility.

41	14. Uncertainty Scenarios	**C**	Analysis is still possible under uncertainty; C is incorrect.
42	14. Uncertainty Scenarios	**D**	Uncertainty = unknown outcomes despite available data; the others are complexity, ambiguity, volatility.
43	15. Unsupervised Learning	**A**	Unsupervised learning discovers structure in unlabeled data with no predefined outputs.
44	15. Unsupervised Learning	**A**	Unsupervised learning needs no labels; A is incorrect.
45	15. Unsupervised Learning	**D**	Grouping unlabeled data by similarity is unsupervised clustering.
46	16. Requisite Variety	**C**	Requisite variety: the controller needs at least as much variety as the system/environment.
47	16. Requisite Variety	**A**	Requisite variety is about control capacity, not raw data volume; A is incorrect.
48	16. Requisite Variety	**B**	If environmental variety exceeds capacity, the environment dominates ('fatal undercomplexity').
49	17. System Dynamics	**C**	System Dynamics models time behaviour using stocks, flows, feedback loops and delays.
50	17. System Dynamics	**C**	System Dynamics is explicitly time-based; C is incorrect.
51	17. System Dynamics	**A**	Stocks accumulate; flows are rates. Balancing loops dampen, reinforcing loops amplify; delays matter.
52	18. Smart Factory	**B**	A smart factory is a CPS-based connected production system - the core of Industry 4.0.
53	18. Smart Factory	**A**	A smart factory is real-time and connected, not purely offline; A is incorrect.
54	18. Smart Factory	**A**	A smart factory is a real-time, connected CPS, not an offline database; A is incorrect.
55	19. Performance Model-Evaluation	**A**	Accuracy can look high while the minority class is entirely missed; recall/precision/F1 are safer.
56	19. Performance Model-Evaluation	**B**	That definition describes precision, not recall; B is incorrect.
57	19. Performance Model-Evaluation	**C**	When false negatives are most costly, recall (finding actual positives) is prioritized.
58	20. Unsupervised Algorithm	**A**	K-means is an unsupervised clustering algorithm; the others are supervised.
59	20. Unsupervised Algorithm	**A**	Fuzzy C-means allows partial (soft) membership; A is incorrect.
60	20. Unsupervised Algorithm	**D**	K-medoids uses a real representative object (medoid) as centre -> robust to outliers.
61	21. Target Centric Approach	**C**	The target-centric approach is structured and iterative, refining insights non-linearly around a goal.
62	21. Target Centric Approach	**B**	Target-centric means goal-first, not data-first; B is incorrect.

63	21. Target Centric Approach	**D**	The target-centric approach is goal-first, not data-first; D is incorrect.
64	22. Clustering & Network Analysis	**A**	Clustering = grouping by similarity; network analysis = studying relationships/connections.
65	22. Clustering & Network Analysis	**C**	Both are unsupervised/exploratory and do not require labels; C is incorrect.
66	22. Clustering & Network Analysis	**C**	Clustering groups by similarity; network analysis studies nodes and the connections between them.
67	23. Hype Myths in Data Science	**B**	'More data is always better' is the myth; relevance and quality matter more than sheer volume.
68	23. Hype Myths in Data Science	**C**	Volume does not beat quality; C is the myth and therefore incorrect.
69	23. Hype Myths in Data Science	**A**	~95% of big data is unstructured and veracity is essential; the others are common myths.
70	24. EDA	**D**	EDA is the early step of understanding the data before modeling.
71	24. EDA	**A**	EDA should not be skipped; A is incorrect.
72	24. EDA	**D**	Using the test set during training is data leakage; D is incorrect.
73	25. Ashby Misinterpretation	**C**	Blindly adding internal complexity is a misinterpretation; attenuation is equally valid and matching variety is correct.
74	25. Ashby Misinterpretation	**A**	Equating variety with data volume is a misinterpretation; A is incorrect.
75	25. Ashby Misinterpretation	**A**	Correct reading = match variety via amplification or attenuation; the others are misinterpretations.
76	26. VUCA environment	**D**	VUCA = Volatility, Uncertainty, Complexity, Ambiguity.
77	26. VUCA environment	**D**	The four VUCA dimensions are distinct; D is incorrect.
78	26. VUCA environment	**D**	Complexity = many interrelated variables; the other three pairings are swapped.
79	27. Type of learner	**D**	KNN is the classic lazy (instance-based) learner; the others are eager learners.
80	27. Type of learner	**C**	KNN is a lazy learner with slow prediction; C is incorrect.
81	27. Type of learner	**B**	It is reversed: eager learners predict fast, lazy learners predict slowly. B is incorrect.
82	28. Biases	**A**	Sampling bias = unrepresentative sample. D=confirmation, B=survivorship, C=automation bias.
83	28. Biases	**B**	Automation tends to scale bias, not remove it; B is incorrect.
84	28. Biases	**D**	Historic inequities in the data are learned and amplified by the model (algorithmic bias).

85	29. VSM Early Warning	**A**	Anti-oscillatory feedback loops detect and dampen emerging instability - an early-warning function.
86	29. VSM Early Warning	**B**	Removing feedback destroys early warning; B is incorrect.
87	29. VSM Early Warning	**A**	Removing feedback destroys early warning, not improves it; A is incorrect.
88	30. Interdisciplinarity in Data Science	**A**	Data Science is transdisciplinary, integrating several fields to solve complex problems.
89	30. Interdisciplinarity in Data Science	**C**	Data Science is transdisciplinary, not purely software engineering; C is incorrect.
90	30. Interdisciplinarity in Data Science	**D**	Data Science genuinely integrates several fields (incl. domain knowledge and social science).